

24CS01EF - Python Full Stack Development – **SDP Document**

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General Instructions

1. These guidelines are provided only to support the development of SDP projects. Students are encouraged to enhance their projects by adding innovative ideas and improvements.
2. All projects **must be developed using the Django framework only**.
3. Each problem statement **can be selected by only one team**.
4. Problem statement selection and allocation will be carried out by the **section faculty**, based on the team's preferences and the availability of problem statements.
5. All teams are instructed to **select the problem statement and begin working immediately**, as this course follows a **purely project-based learning approach**.

List of Problem Statement

PS01 – Smart Agricultural Value-Added Marketplace

Problem Description:

Farmers often depend on middlemen and sell raw agricultural produce at minimal prices, limiting their income. There is a strong need for a digital platform that allows farmers to convert raw crops into value-added products such as packaged food, organic powders, oils, or processed items. This Django-based web application empowers farmers to list such products directly to buyers, ensuring better profit margins. The platform also introduces digital inventory management, order processing, and customer engagement to modernize rural entrepreneurship.

Core Features:

- Product listing and inventory management
- Buyer ordering and order tracking
- Feedback and rating system
- Admin monitoring dashboard

Roles:

Admin, Farmer, Buyer

Description of the Roles:

- **Admin:** Manages users, monitors transactions, and generates system reports
 - **Farmer:** Adds products, updates stock, and processes orders
 - **Buyer:** Browses products, places orders, and provides feedback
-

PS02 – Tribal and Handicraft E-Commerce Platform

Problem Description:

Tribal and rural artisans possess exceptional skills but lack access to digital marketplaces. This project builds a Django-based e-commerce platform that enables artisans to showcase and sell their handcrafted products globally. The system preserves cultural identity while ensuring fair trade practices. By digitizing product listings, orders, and reviews, the platform bridges the gap between artisans and buyers worldwide.

Core Features:

- Product uploads with images
- Order and payment tracking
- Ratings and reviews
- Cultural authenticity validation

Roles:

Admin, Artisan, Buyer, Cultural Consultant

Description of the Roles:

- **Admin:** Oversees platform operations and resolves issues
 - **Artisan:** Uploads and manages handicraft products
 - **Buyer:** Purchases products and provides reviews
 - **Cultural Consultant:** Verifies authenticity and cultural accuracy
-

PS03 – Cultural Heritage and Virtual Tourism Portal

Problem Description:

India's cultural heritage and historical monuments are vast but not easily accessible to everyone. This platform digitizes cultural knowledge through articles, images, and virtual tours. It helps users explore monuments, traditions, and history interactively. Django manages content creation, multimedia storage, and user engagement effectively.

Core Features:

- Virtual tours and galleries
- Cultural articles and multimedia
- User discussions and engagement

Roles:

Admin, Cultural Enthusiast, Content Creator, Tour Guide

Description of the Roles:

- **Admin:** Moderates content and ensures factual accuracy
 - **Cultural Enthusiast:** Explores and interacts with cultural content
 - **Content Creator:** Uploads educational materials
 - **Tour Guide:** Provides expert insights during virtual tours
-

PS04 – Domestic Violence Support and Legal Aid System

Problem Description:

Victims of domestic violence often struggle to access confidential support and accurate legal information. This secure Django-based application provides a safe space to seek help, counseling, and legal guidance. The platform prioritizes privacy, data encryption, and controlled access. It connects survivors with professionals while ensuring anonymity and safety.

Core Features:

- Secure help request system
- Counseling and legal resources
- Confidential communication

Roles:

Admin, Survivor, Counsellor, Legal Advisor

Description of the Roles:

- **Admin:** Maintains platform security and content
 - **Survivor:** Accesses resources and seeks help
 - **Counsellor:** Provides emotional and psychological support
 - **Legal Advisor:** Offers legal advice and assistance
-

PS05 – Emergency Donation and Relief Management System

Problem Description:

During disasters and emergencies, donation efforts often suffer from mismanagement and lack of transparency. This platform organizes donors, recipients, and logistics teams under a single system. Django enables real-time tracking of donations and deliveries. The platform ensures fair distribution and accountability.

Core Features:

- Donation and request listings
- Delivery tracking
- Transparency reports

Roles:

Admin, Donor, Recipient, Logistics Coordinator

Description of the Roles:

- **Admin:** Oversees relief operations and ensures transparency
 - **Donor:** Lists and tracks donated items
 - **Recipient:** Requests and confirms receipt of items
 - **Logistics Coordinator:** Manages transportation and delivery
-

PS06 – Online Learning Management System (LMS)

Problem Description:

Online education requires a structured system for managing courses, students, and assessments. This Django-based LMS enables instructors to create courses and track student progress digitally. Students can access learning materials, submit assignments, and receive grades. The platform enhances remote learning efficiency.

Core Features:

- Course creation and enrollment
- Assignment submission
- Grading and feedback

Roles:

Admin, Instructor, Student

Description of the Roles:

- **Admin:** Configures and manages the LMS
 - **Instructor:** Creates courses and evaluates students
 - **Student:** Learns, submits assignments, and tracks progress
-

PS07 – Educational Institution ERP System

Problem Description:

Educational institutions manage vast academic and administrative data manually, leading to inefficiencies. This ERP system centralizes student records, attendance, grades, and schedules. Django ensures secure, scalable data handling. The system improves transparency and reduces paperwork.

Core Features:

- Student information management
- Attendance and grading
- Timetable scheduling

Roles:

Admin, Teacher, Student, Administrator

Description of the Roles:

- **Admin:** Configures ERP modules
 - **Teacher:** Manages academic records
 - **Student:** Views personal academic data
 - **Administrator:** Oversees institutional operations
-

PS08 – Placement and Internship Management Platform

Problem Description:

Manual tracking of placements and internships is time-consuming and error-prone. This platform digitizes job postings, applications, and placement records. Django enables structured workflows and report generation. It benefits students, employers, and placement officers.

Core Features:

- Job and internship postings
- Application tracking
- Placement reports

Roles:

Admin, Student, Employer, Placement Officer

Description of the Roles:

- **Admin:** Controls system access
 - **Student:** Applies for opportunities
 - **Employer:** Posts job openings
 - **Placement Officer:** Tracks placement outcomes
-

PS09 – Career Guidance and Counseling Platform

Problem Description:

Students often face confusion when choosing career paths. This platform provides assessments, counseling sessions, and expert guidance. Django manages appointment scheduling and personalized recommendations. It helps students make informed career decisions.

Core Features:

- Career assessments
- Counseling appointments
- Personalized recommendations

Roles:

Admin, Student, Counselor

Description of the Roles:

- **Admin:** Manages career resources
 - **Student:** Seeks guidance and assessments
 - **Counselor:** Provides expert advice
-

PS10 – Student Performance Analytics System

Problem Description:

Raw academic scores do not fully reflect student performance trends. This system analyzes marks, attendance, and assessments to generate insights. Django creates visual reports and recommendations. Teachers and students benefit from data-driven feedback.

Core Features:

- Performance analytics
- Visual dashboards
- Improvement suggestions

Roles:

Admin, Teacher, Student

Description of the Roles:

- **Admin:** Maintains analytics system
 - **Teacher:** Reviews performance insights
 - **Student:** Tracks academic progress
-

PS11 – Assignment Submission and Peer Review System

Problem Description:

Traditional assignment systems lack transparency and collaboration. This platform enables digital submissions and peer evaluations. Django handles deadlines, feedback, and grading. It promotes collaborative learning and accountability.

Core Features:

- Online assignment submission
- Peer review mechanism
- Feedback tracking

Roles:

Admin (Teacher), Student

Description of the Roles:

- **Admin (Teacher):** Creates assignments and reviews work
 - **Student:** Submits assignments and reviews peers
-

PS12 – Student Project and Portfolio Management System

Problem Description:

Students need a digital space to manage projects and showcase skills. This platform tracks project milestones and builds portfolios. Django supports feedback and evaluation. It enhances employability and academic documentation.

Core Features:

- Project milestone tracking
- Portfolio creation
- Feedback system

Roles:

Admin, Student

Description of the Roles:

- **Admin:** Reviews and evaluates projects
 - **Student:** Manages projects and portfolios
-

PS13 – Scholarship and Financial Aid Tracking System

Problem Description:

Students often miss scholarships due to scattered information. This platform centralizes scholarship listings and application tracking. Django sends deadline alerts and status updates. It improves accessibility and financial planning.

Core Features:

- Scholarship listings
- Application tracking
- Notifications

Roles:

Admin, Student

Description of the Roles:

- **Admin:** Updates scholarship details
 - **Student:** Applies and tracks status
-

PS14 – Mutual Fund Awareness and Investment Platform

Problem Description:

Many individuals invest without understanding risks and returns. This platform educates users about mutual funds and comparisons. Django ensures secure financial data handling. It supports informed investment decisions.

Core Features:

- Fund comparison
- Educational content
- Risk analysis

Roles:

Admin, Investor, Financial Advisor

Description of the Roles:

- **Admin:** Maintains fund data

- **Investor:** Explores investment options
 - **Financial Advisor:** Provides guidance
-

PS15 – Loan Issuance and Repayment Management System

Problem Description:

Loan processing involves complex calculations and tracking. This system digitizes loan applications, approvals, and repayments. Django ensures accurate EMI calculations and secure record management.

Core Features:

- Loan application management
- EMI calculation
- Payment tracking

Roles:

Admin, Lender, Borrower

Description of the Roles:

- **Admin:** Oversees system operations
 - **Lender:** Issues and manages loans
 - **Borrower:** Applies and repays loans
-

PS16 – Election Monitoring and Civic Engagement System

Problem Description:

Transparent elections strengthen democracy. This platform enables citizens to track election updates and report irregularities. Django ensures secure data handling and reporting. It promotes civic awareness.

Core Features:

- Election updates
- Issue reporting
- Analytics dashboard

Roles:

Admin, Citizen, Election Observer

Description of the Roles:

- **Admin:** Manages election data
 - **Citizen:** Reports issues and tracks updates
 - **Election Observer:** Monitors election fairness
-

PS17 – Smart City Information and Issue Reporting System

Problem Description:

Citizens often struggle to find accurate information about city services and report civic issues. This platform provides details about public services and allows users to report problems like road damage or water issues. Authorities can track and update issue status. Django manages reports and user interactions.

Core Features:

- City service directory

- Issue reporting
- Status tracking

Roles:

Admin, User

Description of the Roles:

- **Admin:** Updates city information
 - **User:** Reports and tracks issues
-

PS18 – Professional Service Hiring Platform

Problem Description:

Finding trustworthy professionals for daily services can be challenging. This platform connects users with verified service providers such as electricians, tutors, or designers. Users can book services and provide reviews. The system builds trust through ratings. Django manages profiles and bookings..

Core Features:

- Professional profiles
- Service booking
- Reviews and ratings

Roles:

Admin, Professional, User

Description of the Roles:

- **Admin:** Oversees platform activities
 - **Professional:** Offers services
 - **User:** Hires professionals
-

PS19 – Indian Constitution Learning Platform

Problem Description:

Many citizens lack awareness of constitutional rights and duties. This platform educates users through structured modules, quizzes, and discussions. It promotes civic responsibility and legal awareness. Educators can contribute content. Django supports interactive learning.

Core Features:

- Learning modules
- Quizzes
- Discussion forums

Roles:

Admin, Educator, Citizen

Description of the Roles:

- **Admin:** Maintains content accuracy
 - **Educator:** Creates educational materials
 - **Citizen:** Learns and participates
-

PS20 – Online Medical Consultation System

Problem Description:

Access to healthcare is limited for many people due to distance or time constraints. This platform enables patients to consult doctors online and receive prescriptions digitally. Medical records are stored securely. The system improves healthcare accessibility. Django ensures data privacy and secure access.

Core Features:

- Appointment booking
- Video consultations
- E-prescriptions

Roles:

Admin, Doctor, Patient

Description of the Roles:

- **Admin:** Ensures data security
 - **Doctor:** Conducts consultations
 - **Patient:** Books and attends appointments
-

PS21 – Nutrition and Diet Analysis System

Problem Description:

Unbalanced diets can cause long-term health problems. This platform allows users to enter dietary information and analyze nutrient intake. It identifies deficiencies and provides recommendations. The system promotes healthy eating habits. Django manages diet analysis and reporting.

Core Features:

- Diet tracking
- Nutrient analysis
- Recommendations

Roles:

Admin, User

Description of the Roles:

- **Admin:** Maintains nutrition data
 - **User:** Tracks diet and health
-

PS22 – Student Mental Health and Wellness Platform

Problem Description:

Student mental health requires timely and confidential support. This platform provides access to counseling resources, wellness programs, and anonymous support. Students can seek help without fear of stigma. Counselors can guide students digitally. Django ensures secure communication.

Core Features:

- Wellness resources
- Counseling sessions
- Anonymous support

Roles:

Admin, Student, Counselor

Description of the Roles:

- **Admin:** Manages wellness services
 - **Student:** Accesses support
 - **Counselor:** Provides guidance
-

PS23 – Sustainable Living Education Platform

Problem Description:

Environmental sustainability requires awareness and behavioral change. This platform educates users about eco-friendly practices and sustainable living. Interactive challenges motivate users to adopt green habits. Progress tracking encourages consistency. Django manages learning content and user engagement.

Core Features:

- Learning modules
- Challenges
- Progress tracking

Roles:

Admin, User

Description of the Roles:

- **Admin:** Updates sustainability content
 - **User:** Learns and applies practices
-

PS24 – Webinar and Online Workshop Management System

Problem Description:

Managing online workshops requires proper scheduling, registration, and tracking. This platform allows organizers to host webinars and workshops efficiently. Participants can register and access session recordings. The system improves event organization. Django manages event workflows.

Core Features:

- Event scheduling
- Registration management
- Session recordings

Roles:

Admin, User

Description of the Roles:

- **Admin:** Organizes events
 - **User:** Participates in sessions
-

PS25 – Extracurricular Activity and Achievement Tracker

Problem Description:

Student achievements beyond academics are often not documented systematically. This

platform tracks participation in clubs, sports, and events. It records awards and certifications. Students can showcase achievements digitally. Django manages structured achievement data.

Core Features:

- Activity registration
- Achievement tracking
- Certificates

Roles:

Admin, Student

Description of the Roles:

- **Admin:** Maintains achievement records
 - **Student:** Views accomplishments
-

PS26 – Certification and Skill Tracking Platform

Problem Description:

Professional certifications need proper tracking for validity and renewals. This platform allows users to upload certificates and receive expiry alerts. It helps users manage skill credentials efficiently. Verification improves authenticity. Django manages alerts and storage.

Core Features:

- Certificate uploads
- Expiry alerts
- Verification

Roles:

Admin, User

Description of the Roles:

- **Admin:** Verifies certifications
 - **User:** Manages skill records
-

PS27 – Academic Research Collaboration Platform

Problem Description:

Research collaboration requires effective coordination and document sharing. This platform enables researchers to manage projects, share files, and communicate. It supports teamwork across institutions. Django ensures secure collaboration and access control.

Core Features:

- Project management
- Document sharing
- Communication tools

Roles:

Admin, Researcher

Description of the Roles:

- **Admin:** Manages research projects
 - **Researcher:** Collaborates and shares work
-

PS28 – Digital Educational Resource Library

Problem Description:

Access to quality educational materials is essential for effective learning. This platform organizes digital resources such as notes, textbooks, and research papers. Users can search and download materials easily. Django manages resource access and categorization.

Core Features:

- Resource uploads
- Search and filters
- Download tracking

Roles:

Admin, User

Description of the Roles:

- **Admin:** Curates resources
 - **User:** Accesses materials
-

PS29 – Virtual Career Fair and Networking Platform

Problem Description:

Physical career fairs limit participation due to location constraints. This platform hosts virtual career fairs where students can interact with employers online. Companies can showcase opportunities and collect resumes. Django manages interactions and event data.

Core Features:

- Virtual booths
- Resume uploads
- Live chat

Roles:

Admin, Student, Employer

Description of the Roles:

- **Admin:** Organizes career fairs
 - **Student:** Explores job opportunities
 - **Employer:** Recruits candidates
-

PS30 – Work-Study and Student Employment Management System

Problem Description:

Work-study programs help students gain experience while supporting their education. This system manages part-time job postings, work hours, and feedback. Students can track work records digitally. Django ensures structured employment management.

Core Features:

- Job postings
- Work-hour tracking
- Supervisor feedback

Roles:

Admin, Student

Description of the Roles:

- **Admin:** Manages work-study programs
- **Student:** Tracks work hours and tasks

Field Study

- Understanding the Problem Statement
- Conducting Survey by interviewing different people (stake holders) & analysing the insights from interview process

LinkedIn Article Posting

You are expected to perform a field study on your Problem Statement to gain insight about the users of your project. User Research is the main objective of the field study. As part of the user research, you are expected to document all your findings into a video Blog/article that should be drafted creatively with photos & videos related to your field study on LinkedIn. Note that the video blog MUST only be drafted on LinkedIn as an article with a combination, photos of you and the users of the Business System, Videos taken during field study, templates that you fill for user research, list of users and their user stories.

To perform User Research, we have documented few tips and templates that are available in the link given below.

<http://bit.ly/userresearchtemplates>

Here is a link to help you understand how to post an article in LinkedIn:

<https://www.linkedin.com/help/linkedin/answer/47538/publish-articles-on-linkedin?lang=en>

Also here are few tips to post an effective article in LinkedIn:

<https://www.inc.com/amy-george/how-to-write-an-attention-grabbing-article-on-linkedin.html>

Summary:

You must conduct a field study related to your Problem Statement by visiting business related to your business system and talking to several users related to the business. You may have to conduct interview, surveys and have an elaborate discussion with them to empathize with them in relation to the business. During this process, ensure you gather the following and document as an effective, empathetic and creative article on LinkedIn. Your LinkedIn article MUST contain:

- i. Project Title, Student Team Details along with Faculty Mentor Details
- ii. About your Project/Business System.
- iii. photos during the interaction, interviews and conducting surveys
- iv. analytics through charts based on the surveys you conduct
- v. video recording of how the users use the business system, how users respond to your interviews and your observations during the field study
- vi. Online research of the business system with appropriate links to showcase as proofs of understanding about the business system.
- vii. fill the customer journey map (templates given above).
- viii. User personas (templates given above)
- ix. Empathy map (templates given above)
- x. Affinity Diagram based on the discussions that happen among the students of each SDP group.

- xi. Your meeting with your faculty mentor must be recorded and included in the article. You may schedule this meeting based on mutually agreed timings.
- xii. The final list of users who will be using your software product that you're planning to build after studying the problem statement.
- xiii. User stories for each user on what do they use the product for. The user stories should be built in such a way that effective utilization of core courses.

All the above must be included and built into an engaging and effective article on LinkedIn with a catchy title. You MUST note that an article published in LinkedIn on your profile helps you gain visibility and builds provisions for technical career opportunities. Hence you MUST take this task seriously and independently. Your article MUST be your own understanding of the complete User Research and should not be copied/plagiarised from your own groupmates or from students of other groups. Any plagiarism in building articles on LinkedIn can cause severe damage to your reputation and may lead to loss of career opportunities. Hence take these as clear and important instructions.

Note: ONLY one member from every Student Team MUST post a LinkedIn Article, remaining students in the same can share the article.

Prototype Design

The prototype you build must be an early replica of what your product will eventually look like. The more detailed your prototype is, you will acquire better scores. Also ensure your SDP Spring Boot Web Application with Micro services MUST be built based on this prototype that you have design ONLY. Some popular wireframing tools like Figma, Adobe XD, Canvas etc. must be used to build the prototype. The output to be shown during Review-1 must be an outcome from such wireframing/prototyping tools.

Reference Links for Prototype Design:

<https://medium.com/@Vincentxia77/12-best-ui-prototyping-tools-for-designers-in-2022-4c646a0c762d>

<https://sagaratechnology.medium.com/free-platforms-to-create-prototyping-for-apps-or-web-design-ca70e902f7e2>

<https://webflow.com/blog/prototyping-tools>

All the above must be included and built into an engaging and effective article on LinkedIn with a catchy title. You MUST note that an article published in LinkedIn on your profile helps you gain visibility and builds provisions for technical career opportunities. Hence you MUST take this task seriously and independently. Your article MUST be your own understanding and should not be copied/plagiarised from your own groupmates or from students of other groups. Any plagiarism in building articles on LinkedIn can cause severe damage to your reputation and may lead to loss of career opportunities. Hence take these as clear and important instructions.

Note: Any one member ONLY from every Student Team MUST post a LinkedIn Article, remaining students in the same can share the article.